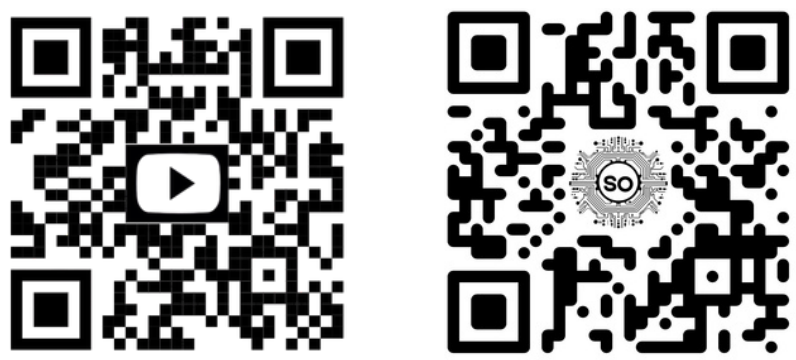




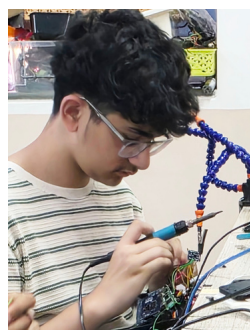
## Robocup Junior Rescue Line

## TEAM

We are System Override, a robotics team that started in 6th grade with EV3 classes. Our early interest in technology turned into a commitment to designing and building autonomous robots. Together, we have created reliable, autonomous, sustainable, open-source solutions that push the boundaries of robotics. Over the years, we have consistently demonstrated excellence on national and international stages, securing 1st place at RoboCup Nationals India (2026), 2nd place at RoboCup Nationals India (2023, 2024, 2025), and 3rd place at RoboCup Regionals India (2024). Internationally, we achieved 1st place in the Primary category at RCAP South Korea(2023), 1st place in the Secondary category at RCAP China(2024), 1st place at the Singapore Open (2025) and the Best TDP award at RCAP Abu Dhabi (2025).



### Ayaan Suri



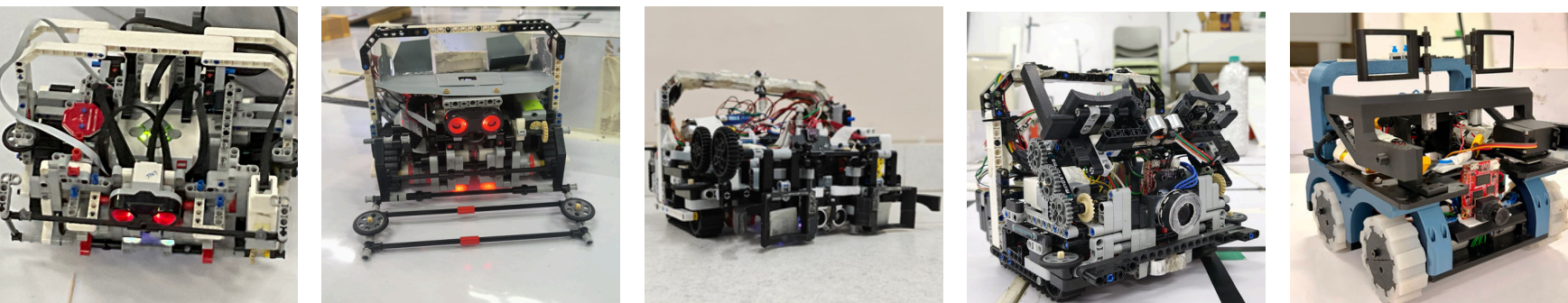
Ayaan, our team lead, has been competing in RoboCup since 2022. He has managed the hardware, electronics, and the robot's controllers. He has trained the AI model and designed the robot in Fusion 360. Beyond technical work, he oversees the team's documentation and helps manage its media presence.

### Tanash Garg



Tanash joined our team in 2025 and is our main programmer. He has helped train the model on Edge Impulse. Additionally, he leads research on innovations for the robot, including exploring and evaluating advanced sensors. He has also contributed to the team's documentation and oversees its media presence.

## ROBOTS JOURNEY



2022

2023

2024

2025

2026

We started with LEGO EV3. In our first year, we built a basic EV3 robot. In our second year, we integrated Arduino with EV3 to help sort victims using Arduino sensors. In our third year, we integrated EV3, Arduino, and an OpenMV camera, using the camera to detect victims and evacuation points. In our fourth year, we kept the same controllers but made the robot lighter and smaller by removing the EV3 casing and adding a longer-lasting battery. In our fifth year, we started building a fully open-source robot using an OpenMV camera for line following and a Raspberry Pi Pico 2 W for managing external sensors.

## HARDWARE

### Microcontrollers

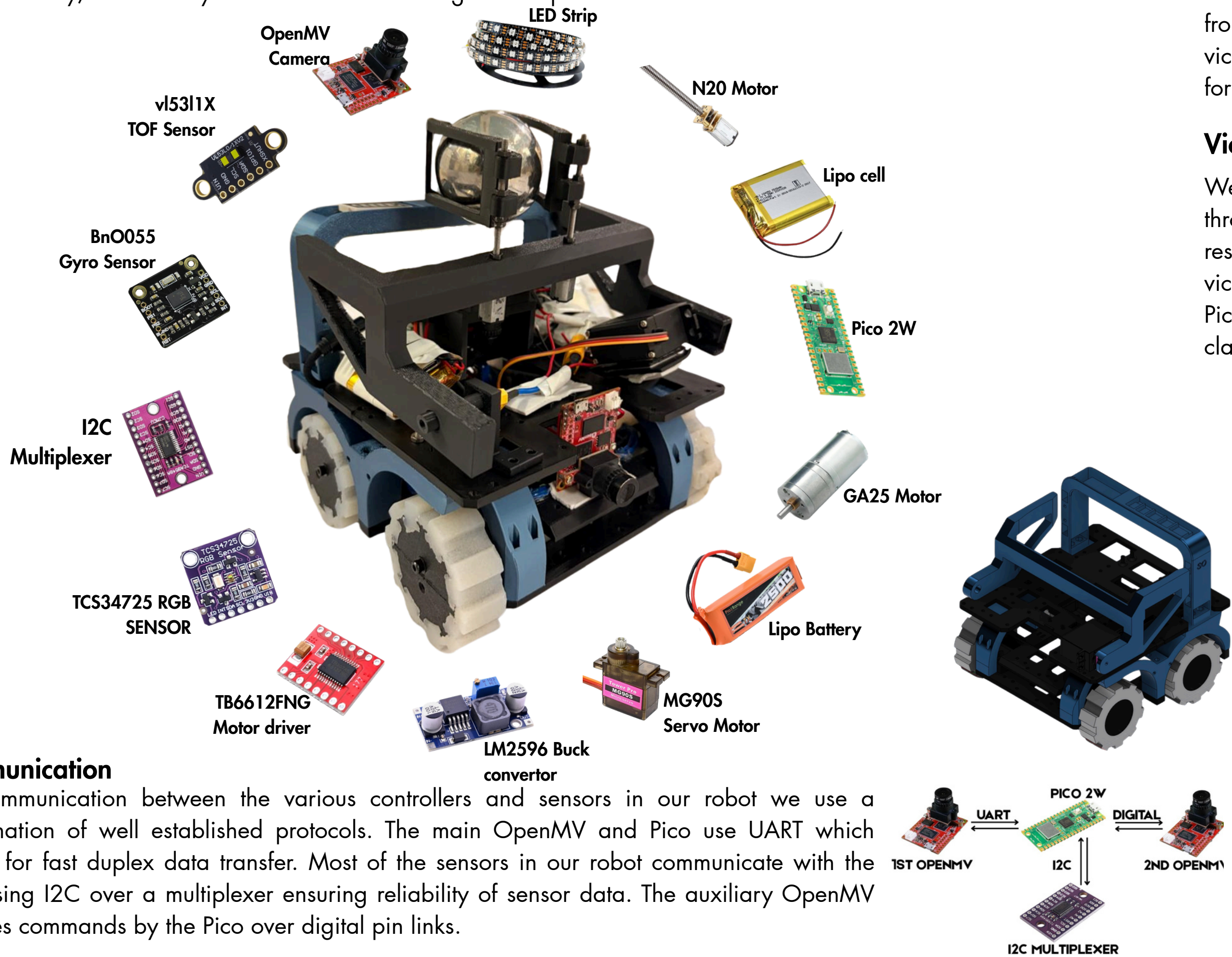
Our main microcontroller is the OpenMV H7 Plus mounted in a downward facing position to detect the line. It has direct control of the drive train motors. The bulk of the sensors and other actuators are controlled by a Raspberry Pi Pico 2W which is subordinate to the OpenMV camera. We also have a secondary OpenMV camera mounted in a forward facing position which acts as a sensor module for rescuing victims.

### Sensors

Due to the use of two cameras, our robot has reduced reliance on other sensors. This allows for greater hardware fault tolerance by virtue of simplicity. We have equipped our robot with a BNO055 9-axis IMU to accurately detect its orientation, 2 VL53L1X ToF distance sensors to detect obstacles and walls, and a TCS34725 RGB sensor to sense colors on the ground. All the sensors communicate with the Pico subcontroller over I2C.

### Innovations

Since our base is entirely custom made we have worked on many ideas which were not possible with the retail kits we used in previous years. These include but are not limited to overlapping the wheel and motor body to save space, custom silicone wheel treads that conform to the speed bump profile, swapping gears in the DC motor gearbox to dial in the required torque/rpm profile, and greater modularity that supports easy assembly/disassembly and maintenance during the competition.



### Communication

For communication between the various controllers and sensors in our robot we use a combination of well established protocols. The main OpenMV and Pico use UART which allows for fast duplex data transfer. Most of the sensors in our robot communicate with the Pico using I2C over a multiplexer ensuring reliability of sensor data. The auxiliary OpenMV receives commands by the Pico over digital pin links.

### Arm

The arm is a simple yet effective design featuring 2 independently actuating claws to grab the victims and a sturdy frame to lift them up. We have used N20 DC Motors for the claws as they are not as sensitive to external load and electrical noise as Servo Motors while remaining just as compact. For lifting the arm we use a single MG995 servo motor that provides sufficient torque and angular control for the task.

### Chasis

Our chassis is made of PLA and manufactured using an FDM 3D printer. After over 14 iterations we have come up with a robust and modular design that utilizes a center mounted camera for line tracking and 4 GA25 6V motors in a tank drive configuration for locomotion. This chassis allows for uniform movement across many ramp and speedbump orientations simplifying the control algorithm.

## SOFTWARE

### Line Following

We use the downward facing OpenMv camera for the lines, greens, silver tape and goal tiles. By using its inbuilt blob detection in 25+ ROIs, we are able to track the positions and shape of all the visible line elements. We combine this information from all the elements into metrics such as slope and deviation to calculate a steering value between -100 and 100 that dictates our bots movement at any instant. Upon detection of greens we execute predetermined turns based on the robots orientation followed by camera based realignment to the black line.

### Evacuation Zone

We enter the evacuation zone by detecting the silver tape using our downward facing OpenMV camera. The RGB sensor is used to recheck silver tape to filter out false positives. Once in the rescue area, we open the arm and move around by selecting a random movement out of several predefined paths as the front facing camera looks for victims. Once a victim is found and picked up, we again use the front camera to search for its corresponding evacuation point and drop it. Once all victims are rescued or the robot has exhausted its allotted rescue time, it searches for the exit by circling the perimeter.

### Victim Detection

We use the front facing OpenMV camera to identify victims in the evacuation zone through custom-trained machine learning models. After successfully entering the rescue area, the camera scans for victims while the robot moves around. Once a victim is detected, the robot approaches, and the camera sends commands to the Pico to pick up the victim using the claw. We also utilise embedded wires in the claw to confirm the presence of live victims.

